

Point (5): the constant $L_{x,\max}$ in Section VIII-A

Diagnosis. With the stated scaling $D_x = \text{diag}(0.01, 0.01, 1, 0.2)$ and the envelope $v \leq 33$ m/s, $|\delta| \leq 0.35$ rad, the induced scaled one-step gain $\|D_x A D_x^{-1}\|$ evaluates to ≈ 1.007 in the infinity norm (and ≈ 1.004 in the spectral norm), confirmed independently by a secant-based Lipschitz estimate of the nonlinear map (≈ 1.006). The value 1.066 is not reproduced by the induced infinity, spectral, or 1-norm of the scaled Jacobian, nor by the raw (unscaled), reversed-scaling, or state-plus-input variants. The contraction conclusion holds in all cases: with 1.007 the margin is in fact better, $\alpha = 1.007 \times 0.80 \approx 0.805 < 1$.

Option A (recommended): report the correct value

Target: the sentence in Section VIII-A giving $L_{x,\max} = 1.066$ and $\alpha = 0.853$. Replace by:

Because the bicycle state mixes meters, radians, and meters per second, the check uses the scaled infinity norm $\|z\|_* = \|D_x z\|_\infty$ with $D_x = \text{diag}(0.01, 0.01, 1, 0.2)$. On the compact envelope $v \leq 33$ m/s and $|\delta| \leq 0.35$ rad, the induced one-step bicycle gain evaluates to $L_{x,\max} = \|D_x A D_x^{-1}\|_\infty \approx 1.007$, dominated by the $\delta \rightarrow \psi$ coupling. Since the accepted anchor mass is lower bounded by $w_0 = \eta_0 \theta_{\min} = 0.20$, Lemma 2 gives $\alpha = L_{x,\max}(1 - w_0) \approx 1.007 \times 0.80 \approx 0.806 < 1$ for the all-to-all accepted-anchor topology.

Option B: keep 1.066 as an explicit conservative bound

Target: same sentence. Keep 1.066 but declare it as an upper bound and add the footnote. Replace by:

Because the bicycle state mixes meters, radians, and meters per second, the check uses the scaled infinity norm $\|z\|_* = \|D_x z\|_\infty$ with $D_x = \text{diag}(0.01, 0.01, 1, 0.2)$. On the compact envelope $v \leq 33$ m/s and $|\delta| \leq 0.35$ rad, we use a conservative one-step bicycle-gain bound $L_{x,\max} = 1.066$.¹ Since the accepted anchor mass is lower bounded by $w_0 = \eta_0 \theta_{\min} = 0.20$, Lemma 2 gives $\alpha = L_{x,\max}(1 - w_0) = 1.066 \times 0.80 \approx 0.853 < 1$ for the all-to-all accepted-anchor topology.

Recommendation. Option A is cleaner and strengthens the margin. Choose Option B only if a single conservative headline constant is preferred; in that case the footnote is needed so a referee reproducing the number is not surprised. [Before submission: confirm from the source script whether 1.066 came from a specific alternative computation, and update the footnote wording accordingly.]

¹The induced scaled-infinity norm $\|D_x A D_x^{-1}\|_\infty$ along the considered trajectory is close to 1.007; the value 1.066 is retained as a conservative upper bound that absorbs Jacobian-evaluation and region margins. Any bound below $1/(1 - w_0) = 1.25$ preserves the one-step contraction.